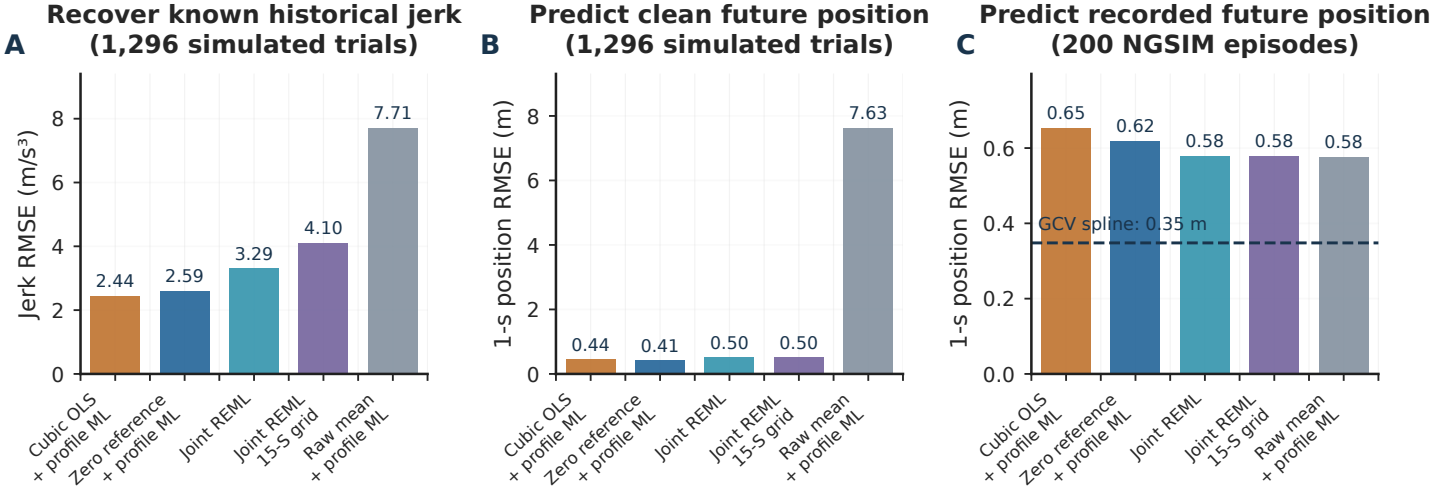
A
Recover known historical jerk
(1,296 simulated trials)
Jerk RMSE (m/s³)
0
2
4
6
8
2.44
2.59
3.29
4.10
7.71
Cubic OLS + profile ML
Zero reference + profile ML
Joint REML
Joint REML 15-S grid
Raw mean + profile ML
B
Predict clean future position
(1,296 simulated trials)
1-s position RMSE (m)
0
2
4
6
8
0.44
0.41
0.50
0.50
7.63
Cubic OLS + profile ML
Zero reference + profile ML
Joint REML
Joint REML 15-S grid
Raw mean + profile ML
C
Predict recorded future position
(200 NGSIM episodes)
1-s position RMSE (m)
0.0
0.2
0.4
0.6
GCV spline: 0.35 m
0.65
0.62
0.58
0.58
0.58
Cubic OLS + profile ML
Zero reference + profile ML
Joint REML
Joint REML 15-S grid
Raw mean + profile ML